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Rigid Body Research under Prof. Baruh

Experiment #1: Free-Fall Validation of a Video Motion Tracker

Model used: Rigid-Body Tracker V5-6

April 20th 2026

Goal

The goal of this experiment was to test whether my MATLAB video tracking system can correctly measure real physical motion. A falling object was used because the motion under gravity is well known and easy to check.

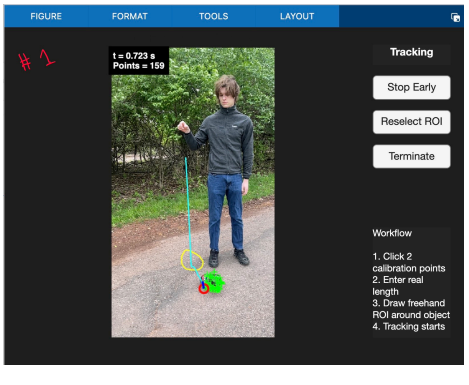
Setup

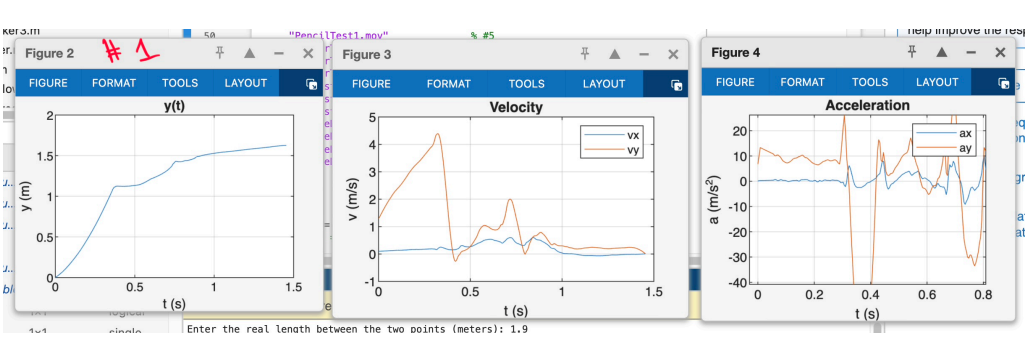
A bottle was dropped by me from about hand height and recorded in slow motion using a phone camera. The video was then processed using the MATLAB tracker.

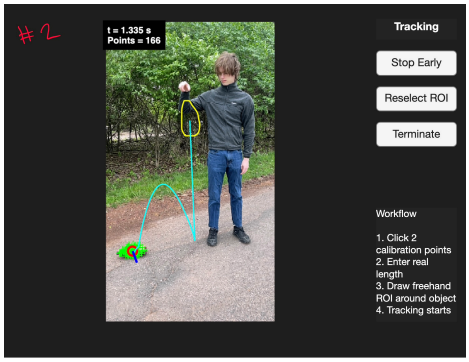
The process was:

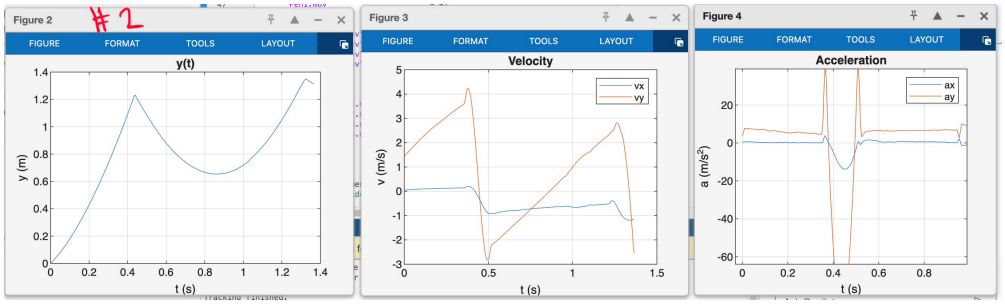
1. Click two points in the video with a known distance to convert pixels to meters.
2. Draw a region of interest around the object.
3. Detect feature points on the object.
4. Track those points through the video.
5. Compute the center position of the object.

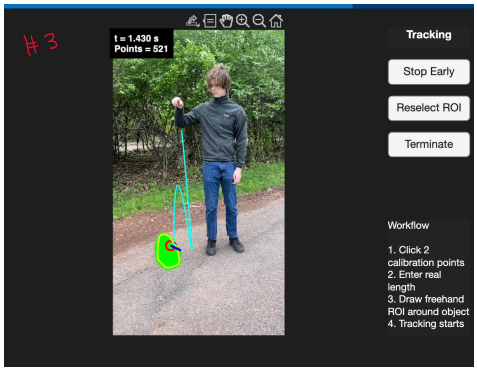
From the tracked position, the program calculates:

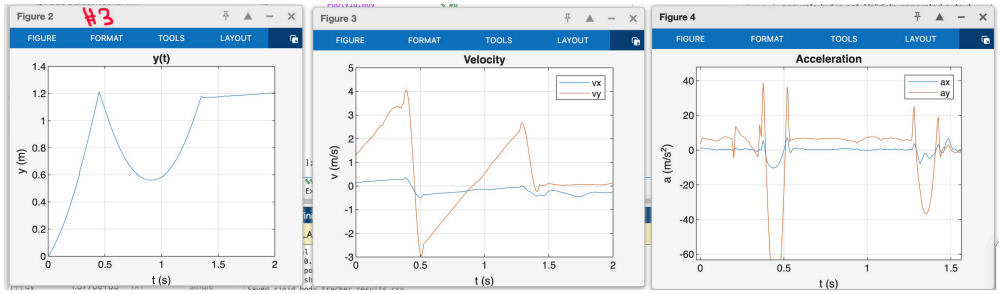


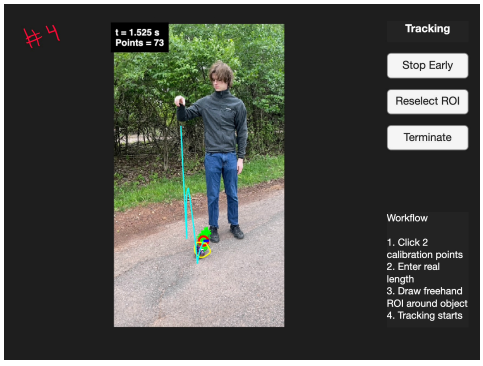


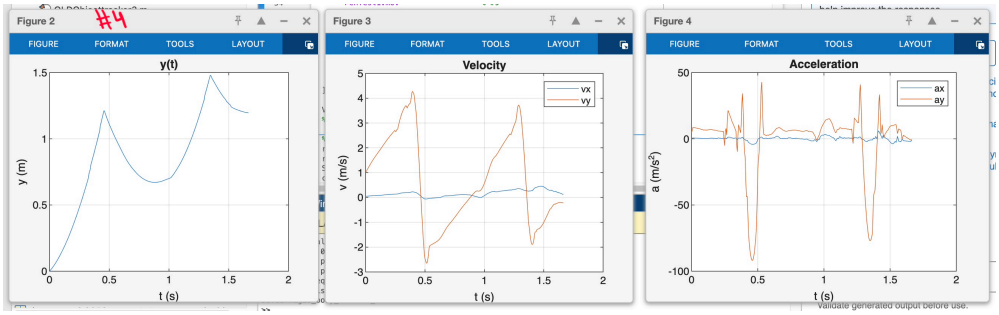












Trial	Velocity Before Impact (m/s)	Velocity After Bounce (m/s)	Coefficient of Restitution
1	~4.4	~0.2	~0.05
2	~4.2	~2.8	~0.67
3	~4.0	~2.6	~0.65
4	~4.3	~2.6	~0.60

- y position y(t)
- velocity v(t)
- acceleration a(t)

It also calculates x-position, theta, angular velocity, angular acceleration, and kinetic/rotational energy, but I did not include those graphs in the report because they are not relevant to this validation experiment

Velocity and acceleration are computed from the position data. A Savitzky–Golay filter was used to derivatives for cleaner data.

Multiple trials were used to check consistency of the measurements.

In the first trial, the bottle hit the ground in a way where there was almost no bounce. It spun a bit then rolled onto the ground. In the other 3, there was a prominent bounce and it made good contact with the ground, as you can see by the parabolic shape the trajectory made.

Results

In all four trials the tracker successfully measured the falling and bouncing motion. In trial 1 the bottle rotated strongly during the impact and the tracker briefly lost the object, so the region of interest had to be reselected.

The position, velocity, and acceleration graphs are consistent with the behavior predicted by kinematics for objects falling under gravity. The position graph follows a parabolic curve, the velocity is approximately linear during the fall, and the acceleration remains roughly constant. The measured acceleration values w 10 m/s^2, which is reasonably close to the expected value of 9.8 m/s^2.

The horizontal velocity stayed close to zero, which matches the fact that the bottle was dropped mostly straight downward.

The coefficient of restitution measurements were also estimated from the velocity graphs. Trial 1 produced almost no rebound because the bottle lost energy to rotation and friction during the impact. The remaining trials produced coefficients of restitution between 0.6 and 0.7, which are consistent with a partially elastic collision.

Conclusion

The results show that the tracking system is able to measure position, velocity, and acceleration from video in a physically reasonable way, along with calculating the coefficient of restitution from impacts. The motion measured during the drop follows the expected behavior for an object moving under gravity.

This experiment suggests that the tracker is accurate enough to analyze simple rigid-body motion from video, and that its results are reasonably accurate. Room for improvement in the center-point tracking is still under progress.